



Service Tool Instruction

Service Tool Instruction
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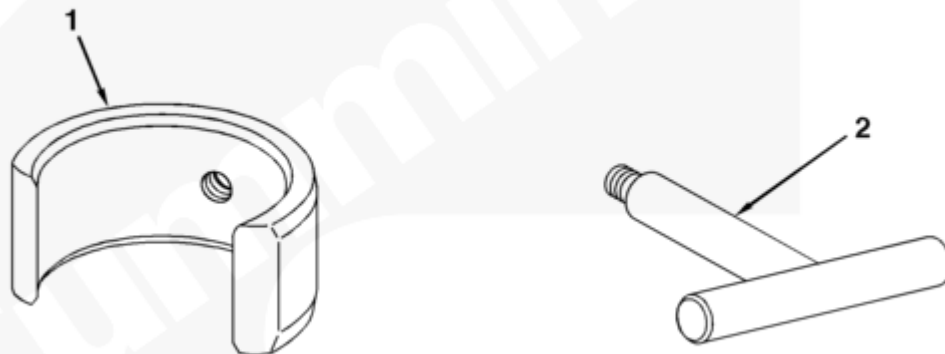
Description

QSK23 and QSK78 Camshaft Removal and Installation Kit

Purpose

This document provides information for using camshaft removal and installation kit, Part Number 3164610. The kit is used to remove and install the camshaft on the QSK23 and QSK78 engines. The camshaft pilot, Part Number 3164845, provides extra stability to the camshaft during the removal and installation procedures, and **must** be used with Part Number 3164610. The long camshaft pilot, Part Number 3164845, can also be used on the QSK45 and QSK60. The short camshaft pilot, Part Number 3163325, can be used on the QSK78 to remove and install the camshaft without removing the flywheel housing.

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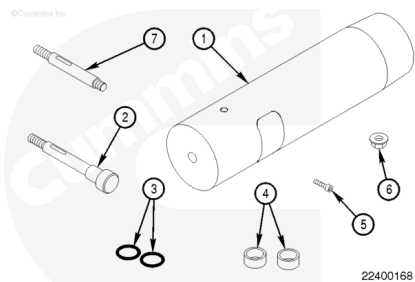
Table 1, QSK23 and QSK78 Camshaft Removal and Installation Kit, Part Number 3164610

Item Number	Part Number	Description	Quantity
1	3164611	Camshaft guide	8
2	3825153	Camshaft guide remover	1

Table 1, QSK23 and QSK78 Camshaft Removal and Installation Kit, Part Number 3164610			
Item Number	Part Number	Description	Quantity
Table 2, Optional Items Used with the Camshaft Removal and Installation Kit, Purchased Separately			
Item Number	Part Number	Description	Quantity
Not shown	3164845	Camshaft pilot kit (long)	1
Not shown	3163325	Camshaft pilot (short)	1

Camshaft Pilot Kit (long), Part Number 3164845

Camshaft Pilot Kit, Part Number 3164845, used with the QSK23 and QSK78 Camshaft Removal and Installation Kit, Part Number 3164610, purchased separately

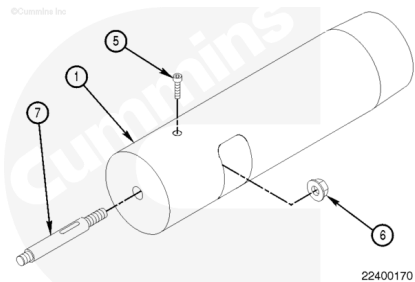


Item	Part Number	Description	Quantity
1	3164701	Camshaft pilot	1
2	3163327	Pilot shaft	1
3	211177	O-ring	2
4	3163323	Spacer	2
5	3077840	Socket head capscrew	1
6	3903210	Nut	1
7	3165109	Pilot shaft	1

QSK23 Engines

Assemble the pilot shaft (7) into the camshaft pilot (1). The end with the smaller threads goes into the pilot with the keyway oriented toward the setscrew hole.

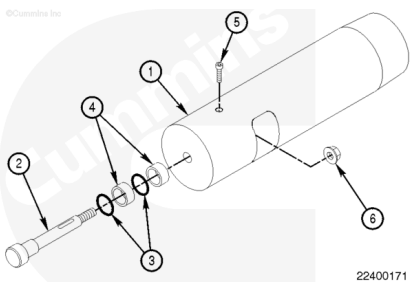
Install the socket head capscrew (5) into the setscrew hole and hand-tighten. The pilot shaft (7) **must** be free to slide back and forth over a short distance. Thread the nut (6) onto the pilot shaft (7) through the opening in the camshaft pilot (1) and do **not** tighten.



QSK78 Engines

Assemble the two o-rings (3) and two spacers (4) onto the pilot shaft (2), as illustrated. Insert the threaded end of the pilot shaft (2) into the camshaft pilot (1) with the keyway oriented toward the setscrew hole.

Install the socket head capscrew (5) into the setscrew hole and hand-tighten. The pilot shaft (2) **must** be free to slide back and forth over a short distance. Thread the nut (6) onto the pilot shaft (2) through the opening in the camshaft pilot (1) and do **not** tighten.

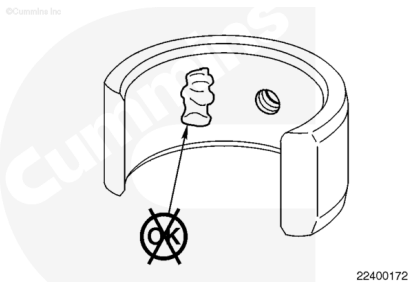


⚠ CAUTION ⚠

Make certain that the camshaft, camshaft guides, and pilot are free of lubricant and debris. Dirt particles on the inside diameter of the camshaft guides and the camshaft lobes during removal can scratch the surface of the camshaft causing performance degradation or damage to the engine.

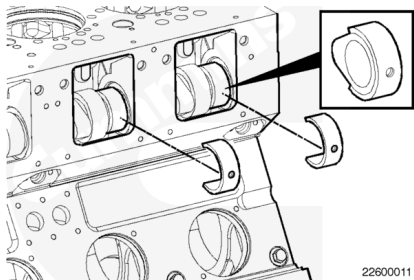
Inspect the camshaft guides for nicks and burrs, and clean thoroughly before each use. Repair or replace damaged camshaft guides.

Do **not** lubricate the inner diameter of the camshaft guides. After installing the camshaft guides, lubricate the outer diameter with Lubriplate™, Part Number 3163086.



The camshaft guides are installed on the inner base circle of the camshaft lobe, **not** the ramp of the camshaft lobe.

Use the chart below to install the camshaft guides on the appropriate exhaust and intake camshaft lobes of the QSK23 or QSK78 engine camshaft.



Camshaft Guide Installation Chart	
QSK78 Camshaft Lobes	
Left Bank	Right Bank
Exhaust - 5, 6, 7, 8, and 9	Exhaust - 4 and 9
Intake - 4 and 9	Intake - 5, 6, 7, 8, and 9

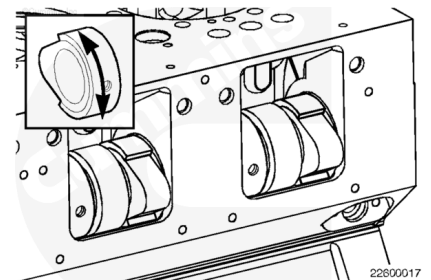
Camshaft Guide Installation Chart	
QSK23 Camshaft Lobes	

Exhaust - 4 and 6
Intake - 1, 3, and 5

Note : The Camshaft Guide Installation chart is for general guidance to facilitate camshaft removal; however, placement of camshaft guides can be changed to suit existing conditions.

It will be necessary to rotate the camshaft to install the camshaft guides. Rotate the camshaft **only** when the camshaft journals are in the camshaft bushings. Do **not** rotate the camshaft when the camshaft guides are in the bushings as this can cause the camshaft to bind.

Adjust the camshaft guides until the camshaft lobe is in the middle of the camshaft guide opening. The camshaft guides **must** be positioned flush with the camshaft lobe.



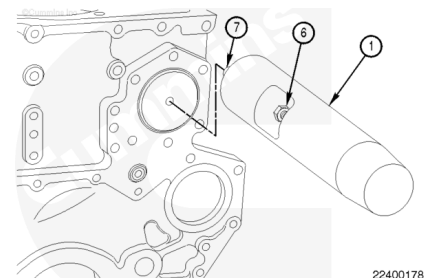
QSK23 Engines

Note : On QSK23 engines, the camshaft pilot is installed on the camshaft at the front of the engine.

Thread the camshaft pilot (1) with the pilot shaft (7) into the threaded camshaft hole. Do **not** tighten.

Slide the camshaft pilot and camshaft assembly rearward until the front camshaft bearing journal and the camshaft pilot are in the front bearing shell. Hand-tighten the nut (6) to secure the camshaft pilot to the camshaft.

The keyway in the camshaft gear **must** be kept at the 6-o'clock position for ease of removal with camshaft guides installed as shown.

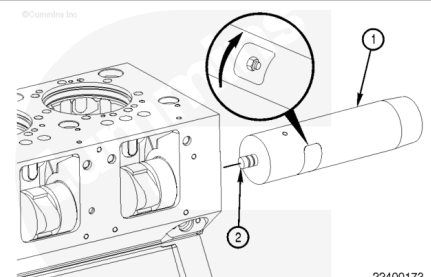


QSK78 Engines

Note : When removing the camshaft with the flywheel housing installed, use short camshaft pilot, Part Number 3163325. If required for assembly, remove the dowel pin from the camshaft pilot mounting face.

Before installing the camshaft pilot (1) onto the camshaft, lightly lubricate the o-ring seals.

Install the camshaft pilot (1) flush against the camshaft. Hand-tighten the nut.



The keyway in the camshaft gear **must** be kept at the 12-o'clock position for ease of removal with camshaft guides installed as shown.

⚠ WARNING ⚠

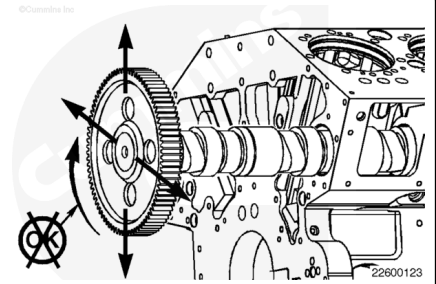
Wear gloves when handling the camshaft. The camshaft has sharp edges that can cause injury if not handled carefully.

⚠ CAUTION ⚠

Do not rotate the camshaft during removal. The camshaft guides can bind and make removal difficult. If necessary, it is permissible to slightly wiggle the camshaft.

Remove the camshaft until it is approximately halfway out of the cylinder block.

As the camshaft is removed, the camshaft guides that have cleared the cylinder block can be moved to other camshaft lobes still within the cylinder block if desired.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

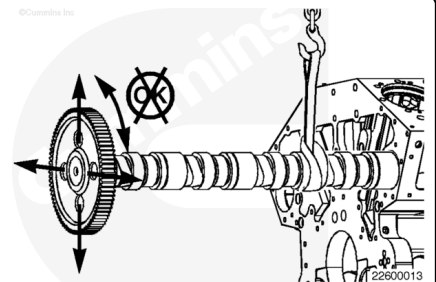
⚠ WARNING ⚠

To reduce the possibility of serious personal injury, stay clear of the camshaft when it is being raised or lowered.

⚠ WARNING ⚠

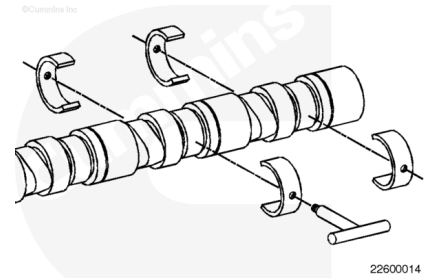
Make sure the camshaft is balanced when lifting. If the camshaft is not balanced, it can fall and cause personal injury.

Use a lifting strap secured around the camshaft to lift the camshaft slightly and support it. Lifting the camshaft too much or too little will bind the camshaft in the bore. Carefully guide the camshaft through the camshaft bore. Make certain that the camshaft is balanced on the lifting strap.

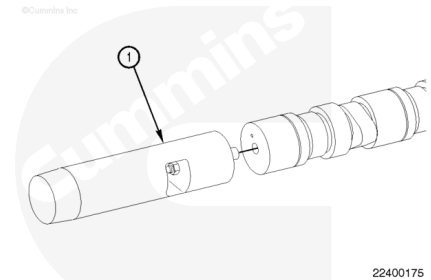


Remove the camshaft from the cylinder block.

After removing the camshaft, use the camshaft guide remover to remove the camshaft guides from the camshaft.



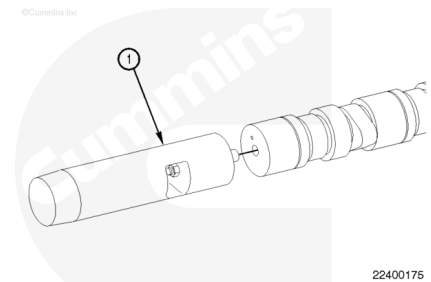
Loosen the nut and remove the camshaft pilot (1) from the camshaft.



QSK23 Engines

Thread the camshaft pilot (1) with pilot shaft (7) into the hole in the end of the camshaft, do **not** tighten.

The keyway in the camshaft gear **must** be kept at the 6 o'clock position for ease of installation with camshaft guides installed as shown.



QSK78 Engines

Note : When installing the camshaft with the flywheel housing installed, use short camshaft pilot, Part Number 3163325. If required for assembly, remove the dowel pin from the camshaft pilot mounting face.

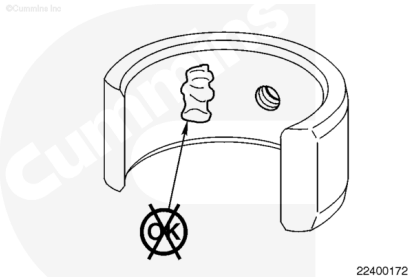
Lubricate the o-rings before installing the camshaft pilot onto the camshaft.

Install the camshaft pilot (1) with the pilot shaft (2) against the camshaft and tighten the nut to the secure assembly.

The keyway in the camshaft gear **must** be kept at the 12 o'clock position for ease of installation with camshaft guides installed as shown.

⚠ CAUTION ⚠

Make certain that the camshaft, camshaft guides, and pilot are free of lubricant and debris. Dirt particles on the inside diameter of the camshaft guides and the camshaft lobes during installation can scratch the surface of the camshaft causing performance degradation or damage to the engine.

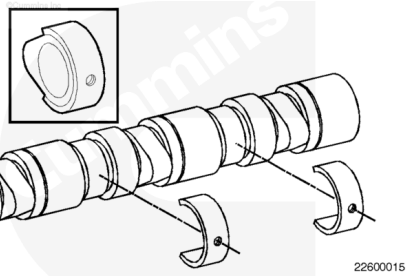


Inspect the camshaft guides for nicks and burrs, and clean thoroughly before each use. Repair or replace damaged camshaft guides.

Do **not** lubricate the inner diameter of the camshaft guides. After installing the camshaft guides, lubricate the outer diameter with Lubriplate™, Part Number 3163086.

The camshaft guides are installed on the inner base circle of the camshaft lobe, **not** the ramp of the camshaft lobe.

Use the chart below to install the camshaft guides on the appropriate exhaust and intake camshaft lobes of the QSK23 or QSK78 engine camshaft.

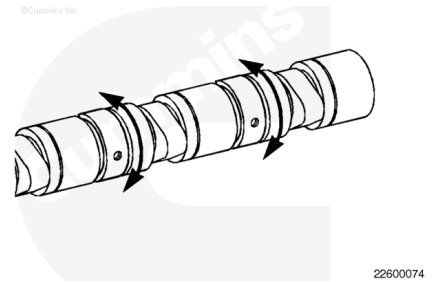


Camshaft Guide Installation Chart	
QSK78 Camshaft Lobes	
Left Bank	Right Bank
Exhaust - 5, 6, 7, 8, and 9	Exhaust - 4 and 9
Intake - 4 and 9	Intake - 5, 6, 7, 8, and 9

Camshaft Guide Installation Chart	
QSK23 Camshaft Lobes	
Exhaust - 4 and 6	
Intake - 1, 3, and 5	

Note : The Camshaft Guide Installation chart is for general guidance to facilitate camshaft installation; however, placement of camshaft guides can be changed to suit existing conditions.

Adjust the camshaft guides until the camshaft lobe is in the middle of the camshaft guide opening. The camshaft guides **must** be positioned flush with the camshaft journal.



⚠ WARNING ⚠

Wear gloves when handling the camshaft. The camshaft has sharp edges that can cause injury if not handled carefully.

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

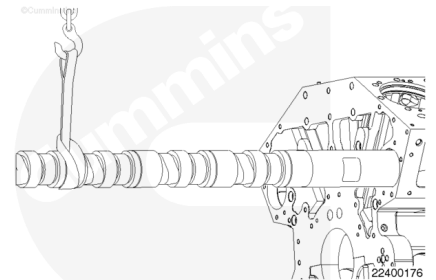
⚠ WARNING ⚠

To reduce the possibility of serious personal injury, stay clear of the camshaft when it is being raised or lowered.

⚠ WARNING ⚠

Make sure the camshaft is balanced when lifting. If the camshaft is not balanced, it can fall and cause personal injury.

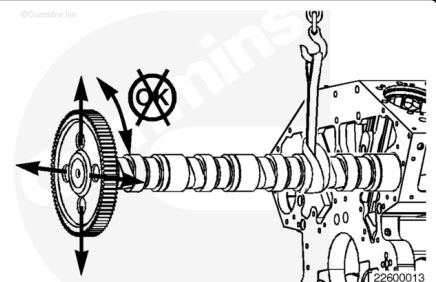
Use a lifting strap secured around the camshaft to lift it into a position parallel to the camshaft bore axis. Make certain that the camshaft is balanced on the lifting strap.



⚠ CAUTION ⚠

Do not rotate the camshaft during installation. The camshaft guides can bind and make installation difficult. If necessary, it is permissible to slightly wiggle the camshaft.

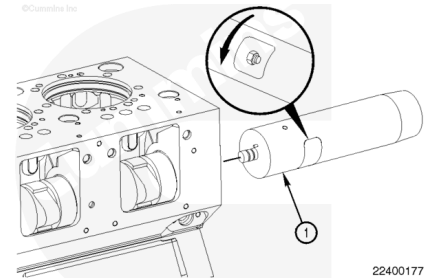
Carefully guide the camshaft into the camshaft bore. Lifting the camshaft too much or too little will bind the camshaft in the bore.



For the QSK23, slide the camshaft pilot and camshaft assembly into the rear bearing until the front bearing journal and the camshaft pilot are in the rear bearing shell. Tighten the nut (6) to secure the pilot to the camshaft.

After the camshaft has been installed halfway, the lifting strap can be removed.

Once the camshaft is installed, loosen the nut and remove the camshaft pilot (1) from the camshaft.



Use the camshaft guide remover to remove the camshaft guides.

It will be necessary to rotate the camshaft to remove the camshaft guides. Rotate the camshaft **only** when the camshaft journals are in the camshaft bushings. Do **not** rotate the camshaft when the camshaft guides are in the bushings as this can cause the camshaft to bind.

